Benchmarking Generative Models for Text-to-Texture Synthesis

Run ID: 20260429-130504-real-diffusion-run

Experiment: real-diffusion-run

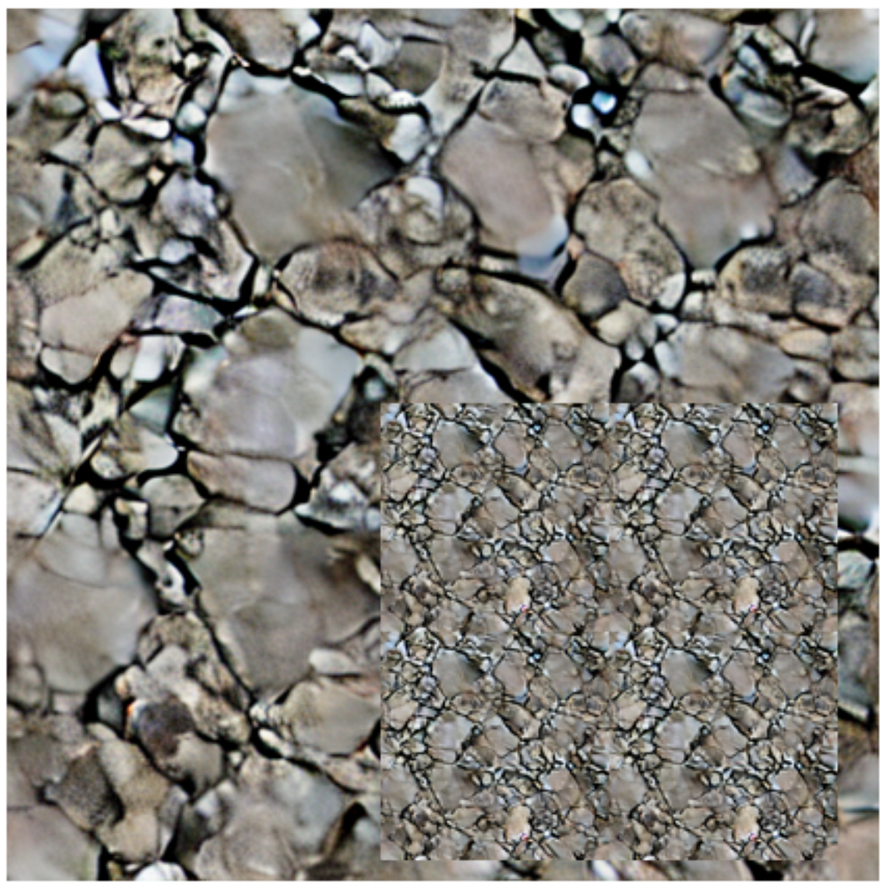
Overall ranking:

stable-diffusion | composite=0.622 | clip=0.298 | seam=0.851 | tiling=0.804 | fft=0.654 | stability=0.784

Generated image examples

wood | sample 0

metal | sample 0



stone | sample 1

Analysis

Mean CLIP alignment is 0.298, mean FFT quality is 0.654, and mean seam continuity is 0.851. Higher CLIP and seam values alongside moderate-to-high FFT values indicate more convincing, production-friendly texture structure.

Across the pilot set, tiling consistency averages 0.804 and seam score averages 0.851. The tiled previews help confirm whether numerical seam quality matches visual repetition behavior.

The widest within-prompt CLIP spread is 0.074 for "seamless stone texture, natural, high detail, seamless texture, PBR material", which makes it the most prompt-sensitive case in this pilot.